SECTION 6.4: WORKING WITH TAYLOR SERIES (BINOMIAL SERIES)

(1) **Starter Definition**

r, n integers

$$\binom{r}{n} = \text{"}r \text{ choose } n\text{"} = \frac{r!}{n!\,(n-r)!} \qquad 0 \le n \le r$$

$$= 0 \qquad \text{otherwise}$$

- Comes from choosing n things from r choices w/o order.
- These are called <u>binomial coefficients</u>

(2) **Starter Examples** Calculate the binomial coefficients below.

(a) $\binom{6}{2} = \dfrac{6!}{2!\,(6-2)!} = \dfrac{6\cdot5\cdot4\cdot3\cdot2\cdot1}{2\cdot1\cdot4\cdot3\cdot2\cdot1} = 15$

(a) $\binom{10}{0} = \dfrac{10!}{0!\,10!} = 1$

(c) $\binom{50}{1} = \dfrac{50!}{1!\,49!} = 50$

(d) $\binom{50}{49} = \dfrac{50!}{49!\,1!}$

(e) $\binom{50}{51} = 0$

(3) Expand $f(x) = (1+x)^6 = (1+x)(1+x)(1+x)(1+x)(1+x)(1+x)$

$$= \underset{\binom{6}{0}}{1} + \underset{\binom{6}{1}}{6x} + \binom{6}{2}x^2 + \binom{6}{3}x^3 + \binom{6}{4}x^4 + \binom{6}{5}x^5 + \underset{\binom{6}{6}}{x^6}$$

This IS the Maclaurin Series!!

(4) Use the definition to find the first ~~four~~ 5 terms of the of a Maclaurin Series for $f(x) = (1+x)^6$.

n	fcn	at a=0
0	$(1+x)^6$	1
1	$6(1+x)^5$	6
2	$6\cdot5(1+x)^4$	$6\cdot5$
3	$6\cdot5\cdot4(1+x)^3$	$6\cdot5\cdot4$

$$1 + 6x + \boxed{\frac{6\cdot5}{2!}}x^2 + \frac{6\cdot5\cdot4}{3!}x^3 + \frac{6\cdot5\cdot4\cdot3}{4!}x^4 + \cdots$$

$$\sum_{n=0}^{\infty}\binom{6}{n}x^n = \sum_{n=0}^{6}\binom{6}{n}x^n$$

b/c all others are zero

(5) **Revisited Definition** $\binom{r}{n} = \dfrac{r!}{n!(n-r)!} = \dfrac{r(r-1)\cdots(r-n+1)}{n!} = \dfrac{\overbrace{r(r-1)\cdots(r-(n-1))}^{r \text{ terms in product}}}{n!}$

Computationally: $\binom{10}{3} = \dfrac{10!}{3!\,7!} = \dfrac{10\cdot 9\cdot 8\cdot 7\cdots 3\cdot 2\cdot 1}{3!\,7\cdot 6\cdots 3\cdot 2\cdot 1}$ We cancel the $(r-n)!$ with part of the $n!$

How do we do this in general?

$$\frac{r(r-1)\cdots(r-n+1)(r-n)\cdots 3\cdot 2\cdot 1}{n!\,(n-r)(n-r-1)\cdots 3\cdot 2\cdot 1} = \frac{r(r-1)(r-2)\cdots(n-r+1)}{n!}$$

THIS formulation makes sense even if r is NOT an integer!

(6) **Harder Examples** Calculate the coefficients below.

(a) $\binom{\frac{1}{2}}{3} = \dfrac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} = \dfrac{3}{8\cdot 3!} = \dfrac{1}{16}$

(a) $\binom{\frac{1}{2}}{0} = 1$

(a) $\binom{\frac{-1}{3}}{2} = \dfrac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)}{2} = \dfrac{2}{9}$

(a) $\binom{3.7}{4} = \dfrac{(3.7)(2.7)(1.7)(0.7)(-0.3)}{4!}$

$= -0.1486$

(7) Use the definition to find the first four terms of the of a Maclaurin Series for $f(x) = (1+x)^{1/2}$.

n	derivative	at $a=0$
0	$(1+x)^{1/2}$	1
1	$\frac{1}{2}(1+x)^{-1/2}$	$\frac{1}{2}$
2	$\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)(1+x)^{-3/2}$	$\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)$
3	$\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(1+x)^{-5/2}$	$\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)$

$$1 + \frac{1}{2}x + \boxed{\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}}x^2 + \boxed{\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}}x^3 + \cdots$$

$$\binom{\frac{1}{2}}{0} \quad \binom{\frac{1}{2}}{1} \quad \boxed{\binom{\frac{1}{2}}{2}}$$

$r = 100$
$n = 4$

$r-(n+1)+1$

$100-5 = 95$

(8) Maclaurin Series for $f(x) = (1+x)^r$, for *any* real number r

$$\sum_{n=0}^{\infty} \binom{r}{n} x^n$$

I.O.C.:

$$\lim_{n\to\infty} |x| \cdot \frac{\binom{r}{n+1}}{\binom{r}{n}} = |x| \lim_{n\to\infty} \left(\frac{r(r-1)\cdots(r-n)}{(n+1)!} \cdot \frac{n!}{r(r-1)\cdots(r-n+1)} \right)$$

$r-(n-1)$

$100-4$
$= 96$

$$= |x| \lim_{n\to\infty} \left| \frac{r-n}{n+1} \right| = |x| \lim_{n\to\infty} \left| \left(\frac{r}{n+1} - \frac{n}{n+1} \right) \right| = |x| < 1$$

(5) **Revisited Definition**

(6) **Harder Examples** Calculate the coefficients below.

(a) $\binom{\frac{1}{2}}{3}$

(a) $\binom{\frac{1}{2}}{0}$

(a) $\binom{\frac{-1}{3}}{2}$

(a) $\binom{3.7}{4}$

(7) Use the definition to find the first four terms of the of a Maclaurin Series for $f(x) = (1+x)^{1/2}$.

(8) Maclaurin Series for $f(x) = (1+x)^r$, for *any* real number r